DEEPFAKE DETECTION

Low-Level Design

API Endpoint Definitions and I/O Specifications

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1. API Endpoint Overview

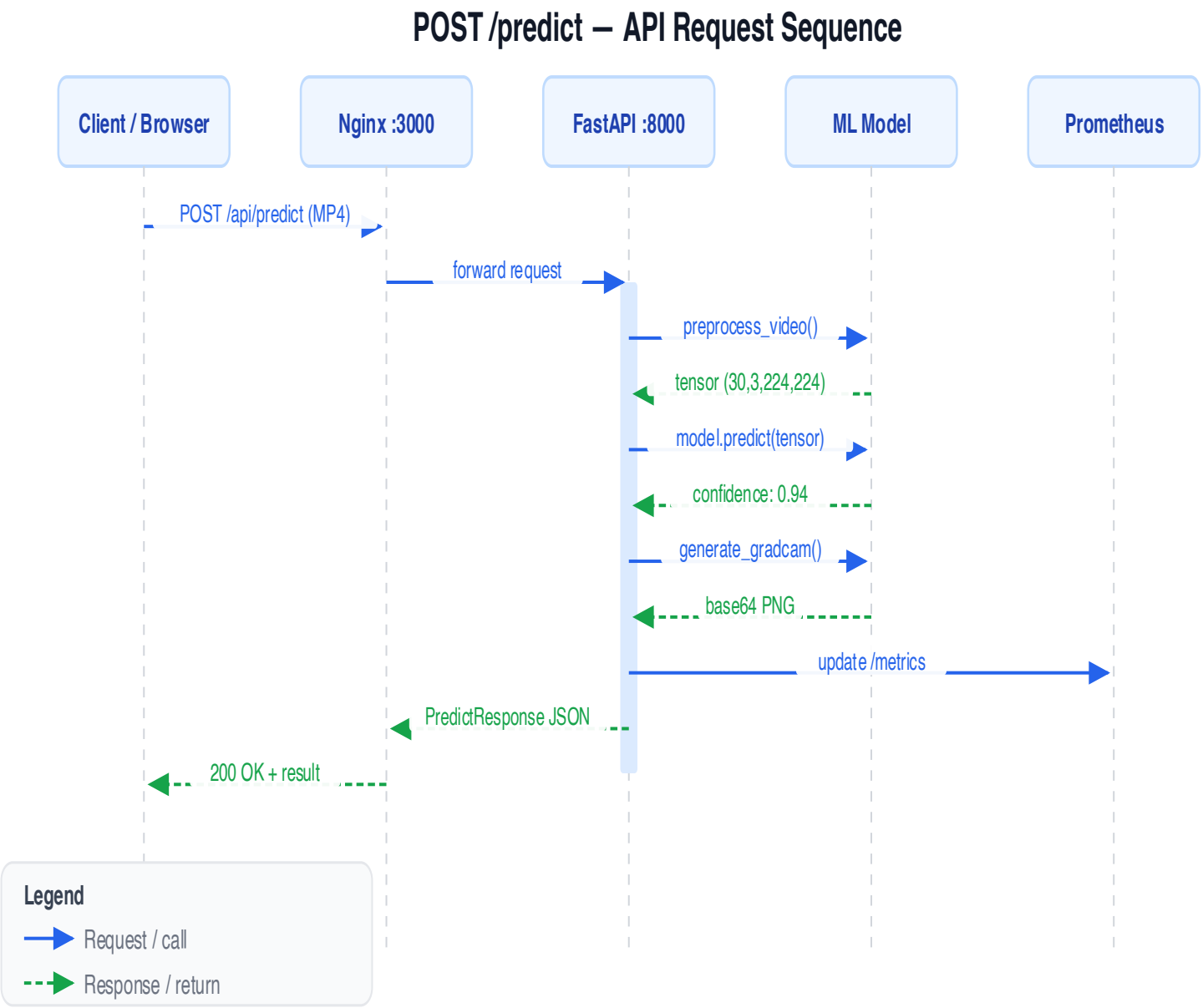

Figure 5: POST /predict — API Request Sequence Diagram

The FastAPI backend exposes a RESTful API. All endpoints accept and return JSON (except /predict which accepts multipart/form-data and /metrics which returns Prometheus text format).

Endpoint	Method	Auth	Purpose
/predict	POST	None	Video deepfake classification
/predict/batch	POST	None	Batch classification (max 10 videos)
/health	GET	None	Liveness probe
/ready	GET	None	Readiness probe (model loaded check)
/metrics	GET	None	Prometheus metrics export
/feedback	POST	None	Submit ground-truth label
/admin/reload-model	POST	Internal only	Force model reload
/admin/rollback	POST	Internal only	Roll back to specific model version
/admin/model-info	GET	Internal only	Current model metadata
/pipeline/mlflow-runs	GET	None	Recent MLflow experiment runs
/pipeline/airflow-runs	GET	None	Recent Airflow DAG runs
/pipeline/throughput	GET	None	Pipeline throughput metric
/support/tickets	POST/GET	X-Username header	Support ticket management
/support/chat	POST	None	Rule-based chatbot

2. Endpoint Specifications

2.1 POST /predict

Classifies a single MP4 video as real or fake.

Request

```
Content-Type: multipart/form-data
field: file (MP4, max 100MB)
```

Success Response — HTTP 200

```
{
  "prediction": "fake",
  "confidence": 0.94,
  "inference_latency_ms": 143.2,
  "gradcam_image": "<base64 PNG string>",
  "mlflow_run_id": "uuid-string",
  "frames_analyzed": 30
}
```

Error Responses

HTTP Code	Condition	Response Body
400	File is not .mp4 extension	{"detail": "Only MP4 files are accepted"}
500	Model inference failure	{"detail": "<error message>"}

Processing Pipeline

1. Save upload to temp file (tempfile.NamedTemporaryFile)
2. preprocess_video(tmp_path) → tensor (30, 3, 224, 224)
3. model.predict(tensor.unsqueeze(0).numpy()) → confidence float
4. prediction = 'fake' if confidence >= 0.5 else 'real'
5. Feature drift check: compute_drift_score(features, baseline)
6. generate_gradcam(model, frames_tensor) → base64 PNG (best-effort)
7. Update Prometheus metrics (request count, latency, confidence, drift)

2.2 POST /predict/batch

Classifies up to 10 MP4 videos in a single request.

Request

```
Content-Type: multipart/form-data
field: files (list of MP4, max 10)
```

Success Response — HTTP 200

```
{
  "results": [
    {"filename": "v1.mp4", "prediction": "fake", "confidence": 0.87,
"inference_latency_ms": 132.1, "error": null},
    {"filename": "v2.mp4", "prediction": null, "confidence": null,
"error": "Only MP4 files are accepted"}
  ],
  "total": 2, "succeeded": 1, "failed": 1
}
```

2.3 GET /health

```
{"status": "ok", "model_loaded": true}
```

2.4 GET /ready

```
{"status": "ready", "model_version": "deepfake/Production/3"}
```

Returns HTTP 503 with {"detail": "Model not loaded"} when model is unavailable.

2.5 GET /metrics

Returns Prometheus text exposition format. Metrics exposed:

Metric Name	Type	Labels	Description
deepfake_requests_total	Counter	method, endpoint, status	Total prediction requests
deepfake_request_latency_seconds	Histogram	endpoint	End-to-end request latency
deepfake_inference_latency_ms	Histogram	—	Model inference time only
deepfake_confidence_score	Histogram	—	Confidence distribution
deepfake_predictions_total	Counter	label (real/fake)	Predictions by class
deepfake_drift_score	Gauge	—	Current feature drift z-score
deepfake_drift_detected_total	Counter	—	Requests with drift detected
pipeline_validation_failures_total	Counter	—	Schema validation failures

2.6 POST /feedback

Request

```
{
  "request_id": "uuid-string",
  "predicted": "fake",
  "ground_truth": "real"
}
```

Validation: predicted and ground_truth must be "real" or "fake" — returns HTTP 422 otherwise.

Response — HTTP 200

```
{"status": "logged", "request_id": "uuid-string"}
```

2.7 POST /admin/rollback

```
Request:  {"version": "2"}
Response: {"status": "reloaded", "model_version": "models:/deepfake/2"}
```

2.8 GET /admin/model-info

```
{"model_version": "models:/deepfake/Production", "run_id": "abc123def456",
"model_loaded": true}
```

2.9 Support Endpoints

Endpoint	Method	Headers	Key Behaviour
/support/tickets	POST	X-Username	Create ticket; 201 with TicketResponse
/support/tickets	GET	X-Role, X-Username	Admin sees all; user sees own tickets
/support/tickets/{id}/resolve	PATCH	X-Role: admin	Set status=resolved; 403 if not admin
/support/chat	POST	—	Rule-based keyword reply; no LLM

3. Module Reference

3.1 backend/app/preprocessing.py

Function	Signature	Description
get_mtcnn()	() → MTCNN	Lazy singleton; device=cpu
extract_frames()	(path, num_frames=30) → list[np.ndarray]	Sample frames evenly via np.linspace; raises ValueError if unreadable
detect_faces()	(frames) → list[np.ndarray]	MTCNN detection; fallback to resized full frame on miss
preprocess_video()	(path, num_frames=30) → torch.Tensor	Full pipeline; pads by repeating last frame if needed; shape (30,3,224,224)

3.2 backend/app/model_loader.py

Symbol	Type	Description
_model	module-level var	Singleton loaded model; None until load_model() called
_current_version	module-level var	Current model URI string
load_model()	() → None	Reads MODEL_NAME/MODEL_STAGE env vars; calls mlflow.pyfunc.load_model
reload_model()	() → None	Force-reloads for rollback support

3.3 backend/app/drift_detector.py

Function	Description
compute_drift_score(features, baseline)	Mean absolute z-score; zero-std guard via np.where(std==0, 1e-8, std)
is_drifted(score, threshold=3.0)	Returns True if score exceeds threshold
load_baseline()	Returns None if ml/feature_baseline.json not found

4. Airflow DAG Reference

4.1 deepfake_pipeline (daily)

```
ingest_videos → extract_frames → detect_faces → compute_features
  → validate_schema → record_baseline_stats → version_with_dvc
```

4.2 retraining_dag (weekly)

```
check_drift (BranchPythonOperator)
  ├── [drift detected]
  │   fetch_new_data → run_mlproject → evaluate_model
  │   → register_if_better → promote_to_production
  └── [no drift]
      skip_retraining (DummyOperator)
```

5. MLflow Model Registry Lifecycle

```
None → Staging → Production → Archived
```

Stage	Trigger	Action
None	Training run completes	Model version created automatically
Staging	register_if_better() — F1 ≥ 0.90	Version transitioned to Staging
Production	promote_to_production()	Staging → Production; current Production → Archived
Archived	Rollback or new promotion	Retained indefinitely for rollback